

Anaya AI 1.2

Feature Guide

A private, mostly-local voice assistant for Windows.

Hold a key, speak, and Anaya controls your computer, answers questions, sets reminders, and types for you.

Developed by	Santosh Pandit
Based on	Maya AI 1.2 by Taha Shaikh (MIT License)
Version	1.2 (Windows port, with macOS support retained)
Repository	github.com/nextginfosoft/Anaya-AI
Language	Python 3.14 (single file: main.py)
Quality	141 automated tests, all passing
Local AI	Ollama (llama3.2:3b) for chat, summaries and translation
Speech	Offline Whisper (base) for recognition; Microsoft neural voice for replies

1. How you use Anaya

- **Push-to-talk (default):** hold **Space** for about a third of a second (or hold **F9**), speak, release. A high beep marks the start of recording and a low beep the end. A quick tap of Space still types a space. Anaya only records while the key is held, so nothing listens in the background.
- **Interrupt:** press **Esc** or a talk key while Anaya is speaking to cut her off.
- **Wake-word mode:** set ANAYA_MODE=wake to say "Anaya, open Chrome" instead of holding a key.
- **Auto-start:** Anaya starts when you sign in to Windows (Startup-folder shortcut, no console window) and logs to anaya.log. Only one copy can run at a time.
- **Follow-ups:** after an answer you can ask a related question; Anaya remembers the last 6 exchanges.

Example

You say	Anaya does
"Open Chrome"	Opens Google Chrome and says "Opening Chrome".
"Remind me in ten minutes to call Raj"	Stores the reminder and speaks it when it is due.
"Who wrote Hamlet" ... "when was he born"	Answers, then uses the previous answer to understand "he".

2. Everything Anaya can do

Computer control

Say	What happens
Open Chrome / VS Code / WhatsApp / YouTube	Launches the app (or its website if the app is missing).
Open Notepad / Calculator / Task Manager / Settings / File Explorer / Paint / Terminal	Opens the Windows tool.
Open folder downloads	Finds and opens the folder by name.
Open github, Open my project, ...	Anything you list in apps.json (URL, folder or program). Edits apply immediately.
Volume up / down / mute / set volume to 40	Changes the system volume (steps of about 10 percent).
Brightness up / down / set brightness to 30	Changes the screen brightness.
Lock the screen	Locks Windows.
Screenshot	Saves a screenshot to the screenshots folder and opens it.
What time is it / what is the date / battery level	Answers out loud.

Web and information

Say	What happens
Search Google for ... / Search YouTube for ...	Opens the search in your browser.
Play Believer	Plays the song on YouTube.
Weather in Delhi / weather	Current temperature and sky (Open-Meteo). Without a city it uses a guess from your IP, or ANAYA_CITY.
Tell me the news	Reads the top 3 headlines (Google News).
Who is ... / Tell me about ...	Two-sentence Wikipedia summary; falls back to the AI if there is no page.

Reminders, timers and briefing

Say	What happens
Remind me in 10 minutes to call Raj	Speaks "Reminder: call Raj" when due. Also works at a clock time ("at 6 pm").
Set a timer for 5 minutes	Speaks "Your timer is done".
What are my reminders / cancel all reminders	Lists or clears them. Reminders are saved to disk and survive restarts.
Give me my briefing	Time, battery, weather, reminders and top headlines. Also plays automatically after sign-in (max once every 4 hours, 12 s after start). Esc stops it.

Hands-free typing and reading

Say	What happens
Start dictation ... Stop dictation	Everything you say is typed into the window you are in. Say "comma", "full stop", "question mark", "new line", "new paragraph" for punctuation. Sentences are capitalised.
Type hello comma world	Types just that one phrase.
Read the clipboard / Read this	Reads the copied or selected text aloud (the clipboard is restored afterwards).
Translate this to Hindi / the clipboard to English	Uses the local model. English is spoken; other languages (Hindi, Marathi, Gujarati, Tamil, Telugu, Bengali, Spanish, French, German) are put on the clipboard.
Summarise what I copied / Summarise this	Two-sentence spoken summary from the local model.

AI chat and memory

- Anything Anaya does not recognise as a command goes to the local model (llama3.2:3b via Ollama).
- Answers are limited to about two spoken sentences (ANAYA_MAX_TOKENS, default 100). Markdown and half-finished sentences are removed before speaking.
- Memory: the last 6 exchanges; resets after 5 quiet minutes. Say "forget that" or "new topic" to clear it.
- Anaya introduces herself when asked "who are you".

Languages

- **English (Indian accent)** and **Hindi**. Anaya picks the language of each recording automatically.
- Say "switch to Hindi" or "switch to English" to choose which is tried first.
- Core commands also work in Hindi and Hinglish (open Chrome, volume up/down, time, date, weather, news).
- Replies are always spoken in English. Hindi text (for example in dictation) is typed correctly.

3. Speech and privacy

Part	How it works	Where your data goes
Hearing	Whisper (base model, about 150 MB) runs on this PC with a vocabulary hint and a filter that removes its usual hallucinations on noise. On quiet test audio it scored 10/10 against Google 8/10, at about 2 s per phrase.	Nowhere. Google is used only if Whisper hears nothing usable (ANAYA_STT_FALLBACK=0 forbids it).
Speaking	Microsoft neural voice (Neerja, Indian English; Swara for Hindi text). Falls back to the offline Windows voice with no internet.	The text Anaya says goes to Microsoft. ANAYA_TTS=windows keeps everything offline.
Private text	Clipboard, selected text, summaries and translations are always spoken with the offline voice.	Stays on this PC.
Thinking	Ollama runs the model locally.	Stays on this PC.
Recordings	Not stored. ANAYA_DEBUG_AUDIO=1 keeps the last 30 rejected clips for debugging.	Local files only.

Lookups for weather, news and Wikipedia send the words of your request to those services.

4. Reliability

- **Single instance:** a second copy exits immediately, so Anaya never answers twice.
- **Crash recovery:** after a crash Anaya restarts with growing delays; after 5 crashes in 10 minutes she stops and says so.
- **Plain-language problems:** she tells you when Ollama is not running, when the model is not installed, when the microphone is unavailable, or when the speech service cannot be reached.
- **Log rotation:** anaya.log is rotated at 1 MB.
- **Missing internet:** speech falls back to the offline voice, and web lookups say they cannot reach the internet.

5. Settings (environment variables)

Variable	Default	Purpose
ANAYA_MODE	ptt	ptt = hold-to-talk, wake = say "Anaya".
ANAYA_PTT_KEY	smart space,f9	Talk keys (comma separated). smart space = hold Space; right ctrl, f8 etc. also work; mouse:x / mouse:x2 for mouse side buttons.
ANAYA_SPACE_HOLD	0.30	Seconds Space must be held before it counts as talking.
ANAYA_STOP_KEY	esc	Key that interrupts speech.
ANAYA_STT / ANAYA_STT_FALLBACK	whisper / 1	Recognition engine; 0 = never use Google.
ANAYA_WHISPER_MODEL	base	Larger models (small, medium) are more accurate but slower.
ANAYA_TTS	edge	edge = neural voice, windows = offline voice.
ANAYA_EDGE_VOICE / _HI	en-IN-NeerjaNeural / hi-IN-SwaraNeural	Neural voices.
ANAYA_VOICE	(auto, female)	Windows offline voice name.
ANAYA_MODEL	llama3.2:3b	Ollama chat model.
ANAYA_MAX_TOKENS	100	Length cap for spoken answers.
ANAYA_CITY	(from IP)	Default city for weather and the briefing.
ANAYA_LANG	en-IN	Language tried first.
ANAYA_BRIEFING / _DELAY	1 / 12	0 turns off the sign-in briefing; delay in seconds.
ANAYA_ANIMATION	0	1 shows the startup GIF.
ANAYA_DEBUG_AUDIO	0	1 keeps rejected recordings.
ANAYA_ALLOW_MULTIPLE	0	1 disables the single-instance guard.

6. How it is built

Layer	Detail
Input	sounddevice records while the key is held; audio is boosted up to 40x for the quiet laptop microphone.
Recognition	Whisper first, then Google (English, then Hindi). Language chosen from Whisper's probabilities.
Command flow	Hindi translation, normalisation, then handlers in order: text tools, briefing, memory reset, reminders, live info, apps.json, system commands, built-in apps, AI chat.
Output	Neural voice via Edge TTS played with Windows MCI (interruptible); offline fallback through System.Speech.
Files	main.py, apps.json, reminders.json, last_briefing.json, anaya.log, requirements.txt, tests/, docs/
Tests	141 pytest tests covering commands, Hindi, reminders, live info (network mocked), dictation, Whisper routing, voice routing and robustness.

7. Known limits and roadmap

Known limits

- The laptop microphone is very quiet. Raising the Windows input level or using a headset improves accuracy most.
- Replies are spoken in English only; Hindi open-ended questions go to a small English-centred model.
- Weather without a city uses an IP-based guess.
- Reminders fire only while Anaya is running.
- Whisper recognition was verified on synthetic quiet speech; live results depend on your voice and room.

Not built yet

- Voice shutdown, restart and sleep (needs a spoken confirmation step to avoid misheard commands).
- System-tray icon with start, stop and quiet mode.
- Calendar, email, music and WhatsApp-sending integrations.
- Hindi spoken replies from the AI.